

Huaibo Huang

Associate Professor
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Education

Institute of Automation, CAS, majors in Computer Application Technology, Advised by Prof. Tieniu Tan	Ph.D. (2019)
Beihang University, majors in Optical Engineering, Advised by Prof. Di Feng	M.E. (2016)
Xi'an Jiaotong University, majors in Measurement & Control Technology and Instruments	B.E. (2012)

Academic Experience

Research Field: Biometrics, Multimodality Intelligence, Generative Learning, and Low-level Vision

Associate Professor at Institute of Automation, Chinese Academy of Sciences	2021 – Present
Assistant Professor at Institute of Automation, Chinese Academy of Sciences	2019 – 2021

Award

First Prize of Wu Wenjun AI Technology Invention Award, awarded by Chinese Association for Artificial Intelligence	2024
Beijing Nova Program, awarded by Beijing Association for Science and Technology	2023
Runner-Up Award in NTIRE 2023 Challenge on 360° Omnidirectional Image Super-Resolution	2023
2nd Place Award in NTIRE 2023 Challenge on Image Super-Resolution (x4)	2023
Member of CAS Youth Innovation Promotion Association, awarded by Chinese Academy of Sciences	2022
BAST Young Talents Lift Engineering, awarded by Beijing Association for Science and Technology	2020
Best Student Paper Award from ITMC Workshops in IEEE International Conference on Multimedia and Expo	2019
Beijing Outstanding Graduates, awarded by Beijing Municipal Education Commission	2019
Chinese Academy of Sciences Presidential Scholarship, awarded by Chinese Academy of Sciences	2019

Services and Societies

Associate Editor, IEEE Transactions on Biometrics, Behavior, and Identity Science (TBIOM)	2024 – present
Area Chair, International Conference on Learning Representations (ICLR)	2026
Area Chair, Neural Information Processing Systems (NeurIPS)	2025 & 2024
Area Chair, ACM International Conference on Multimedia (ACMMM)	2025
Reviewer for IEEE Transactions on Image Processing (TIP)	2021 – present
Reviewer for IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)	2022 – present
Reviewer for International Journal of Computer Vision (IJCV)	2023 – present
IEEE Member	2018 – present

Publications

Google Scholar: 3,600 + Citations, H-index: 27

Published 2 papers in TPAMI, 7 papers in IJCV, 3 papers in TIFS, and 22 papers in NeurIPS/CVPR/ICLR/ICCV.

Selected Publications

1. **Huaibo Huang**, Mandi Luo, Ran He. Memory Uncertainty Learning for Real-World Single Image Deraining. IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**), 45(3): 3446-3460, 2023.
2. **Huaibo Huang**, Aijing Yu, Zhenhua Chai, Ran He, Tieniu Tan. Selective Wavelet Attention Learning for Single Image Deraining. International Journal of Computer Vision (**IJCV**), 129(4): 1282-1300, 2021.
3. **Huaibo Huang**, Ran He, Zhenan Sun, Tieniu Tan. Wavelet Domain Generative Adversarial Network for Multi-scale Face Hallucination. International Journal of Computer Vision (**IJCV**), 127(6-7): 763-784, 2019.
4. **Huaibo Huang**, Xiaoqiang Zhou, Ran He. Orthogonal Transformer: An Efficient Vision Transformer Backbone with Token Orthogonalization. Neural Information Processing Systems (**NeurIPS**), 2022.
5. **Huaibo Huang**, Aijing Yu, Ran He. Memory Oriented Transfer Learning for Semi-Supervised Image Deraining. Computer Vision and Pattern Recognition (**CVPR**), 2021.
6. **Huaibo Huang**, Zhihang Li, Ran He, Zhenan Sun, Tieniu Tan. IntroVAE: Introspective Variational Autoencoders for Photographic Image Synthesis. Neural Information Processing Systems (**NeurIPS**), 2018: 52-63.
7. **Huaibo Huang**, Ran He, Zhenan Sun, Tieniu Tan. Wavelet-SRNet: A Wavelet-based CNN for Multi-scale Face Super Resolution. International Conference on Computer Vision (**ICCV**), 2017: 1698-1706.

Book

Yi Li, **Huaibo Huang**, Ran He, Tieniu Tan. Heterogeneous Facial Analysis and Synthesis. Springer Briefs in Computer Science, Springer 2020, ISBN 978-981-13-9147-7.

Other Refereed Journal Publications

1. Junxian Duan, Hao Sun, Fan Ji, Kai Zhou, Zhiyong Wang, **Huaibo Huang**, Lianwen Jin. RealDTT: Towards A Comprehensive Real-World Dataset for Tampered Text Detection. International Journal of Computer Vision (**IJCV**), 2025.
2. Junxian Duan, Yang Ai, Jipeng Liu, Shenyuan Huang, **Huaibo Huang**, Jie Cao, Ran He. Test-time Forgery Detection with Spatial-Frequency Prompt Learning. International Journal of Computer Vision (**IJCV**), 2024.
3. Xiaoqiang Zhou, Chaoyou Fu, **Huaibo Huang**, Ran He. Dynamic Graph Memory Bank for Video Inpainting. IEEE Transactions on Circuits and Systems for Video Technology (**TCSVT**), 2024.
4. Xiaoqiang Zhou, **Huaibo Huang**, Zilei Wang, Ran He. RISTRA: Recursive Image Super-resolution Transformer with Relativistic Assessment. IEEE Transactions on Multimedia (**TMM**), 2024.
5. Jianze Wei, **Huaibo Huang**, Yunlong Wang, Ran He, Zhenan Sun. Towards More Discriminative and Robust Iris Recognition by Learning Uncertain Factors. IEEE Transactions on Information Forensics & Security (**TIFS**), 2022.
6. Jianze Wei, Yunlong Wang, **Huaibo Huang**, Ran He, Zhenan Sun, Xingyu Gao. Contextual Measures for Iris Recognition. IEEE Transactions on Information Forensics & Security (**TIFS**), 2022.
7. Mandi Luo, Haoxue Wu, **Huaibo Huang**, Weizan He, Ran He. Memory-Modulated Transformer Network for Heterogeneous Face Recognition. IEEE Transactions on Information Forensics & Security (**TIFS**), 2022.
8. Chaoyou Fu, Xiang Wu, Yibo Hu, **Huaibo Huang**, Ran He. DVG-Face: Dual Variational Generation for Heterogeneous Face Recognition. Trans. Pattern Analysis and Machine Intelligence (**TPAMI**), 2021.
9. Aijing Yu, Haoxue Wu, **Huaibo Huang**, Zhen Lei, Ran He. LAMP-HQ: A Large-Scale Multi-Pose High-Quality Database for NIR-VIS Face Recognition. International Journal of Computer Vision (**IJCV**), 2021.
10. Xin Ma, Xiaoqiang Zhou, **Huaibo Huang**, Gengyun Jia, Zhenhua Chai, Xiaolin Wei. Contrastive Attention Network with the Dense Field Estimation for Face Completion. Pattern Recognition (**PR**), 2021.
11. Yi Li, **Huaibo Huang**, Jie Cao, Ran He, Tieniu Tan. Disentangled Representation Learning of Makeup Portraits in the Wild. International Journal of Computer Vision (**IJCV**), 2020, 128: 2166-2184. (Co-first author)
12. Xin Zheng, Yanqing Guo, **Huaibo Huang**, Yi Li, Ran He. A Survey to Deep Facial Attribute Analysis. International Journal of Computer Vision (**IJCV**), 2020.
13. Xin Zheng, **Huaibo Huang**, Yanqing Guo, Ran He. BLAN: Bi-directional Ladder Attentive Network for Facial Attribute Prediction. Pattern Recognition (**PR**), 2020.

Other Refereed Conference Publications

1. Qihang Fan, **Huaibo Huang**, Yang Ai, Ran He. Rectifying Magnitude Neglect in Linear Attention. International Conference on Computer Vision (**ICCV**), 2025. (Highlight, Accepted)
2. Qihang Fan, **Huaibo Huang**, Mingrui Chen, Ran He. Semantic Equitable Clustering: A Simple and Effective Strategy for Clustering Vision Tokens. International Conference on Computer Vision (**ICCV**), 2025. (Accepted)
3. Qihang Fan, **Huaibo Huang**, Ran He. Breaking the Low-Rank Dilemma of Linear Attention. Computer Vision and Pattern Recognition (**CVPR**), 2025.
4. Xuannan Liu, Zekun Li, Pei Pei Li, **Huaibo Huang**, Shuhan Xia, Xing Cui, Linzhi Huang, Weihong Deng, Zhaofeng He. MMFakeBench: A Mixed-Source Multimodal Misinformation Detection Benchmark for LVLMS. International Conference on Learning Representations (**ICLR**), 2025.
5. Yuguang Zhang, Qihang Fan, **Huaibo Huang**. Vision Transformer with Sparse Scan Prior. ACM International Conference on Multimedia (**ACM MM**), 2025.
6. Xiaotian Han, Yiren Jian, Xuefeng Hu, Haogeng Liu, Yiqi Wang, Qihang Fan, Yang Ai, **Huaibo Huang**, Ran He, Zhenheng Yang, Quanzeng You. InfiMM-WebMath-40B: Advancing Multimodal Pre-Training for Enhanced Mathematical Reasoning. Empirical Methods in Natural Language Processing (**EMNLP**), 2025.
7. Yang Ai, Xiaoqiang Zhou, **Huaibo Huang**, Xiaotian Han, Zhengyu Chen, Quanzeng You, Hongxia Yang. DreamClear: High-Capacity Real-World Image Restoration with Privacy-Safe Dataset Curation. Neural Information Processing Systems (**NeurIPS**), 2024.
8. Haogeng Liu, Quanzeng You, Xiaotian Han, Yongfei Liu, **Huaibo Huang**, Ran He, Hongxia Yang. Visual Anchors Are Strong Information Aggregators For Multimodal Large Language Model. Neural Information Processing Systems (**NeurIPS**), 2024.
9. Hongbo Wang, Jin Liu, Xiaoqiang Zhou, Jie Cao, **Huaibo Huang**, Ran He. Hallo3D: Multi-Modal Hallucination Detection and Mitigation for Consistent 3D Content Generation. Neural Information Processing Systems (**NeurIPS**), 2024.
10. Yang Ai, **Huaibo Huang**, Xiaoqiang Zhou, Jiexiang Wang, Ran He. Multimodal Prompt Perceiver: Empower Adaptiveness, Generalizability and Fidelity for All-in-One Image Restoration. Computer Vision and Pattern Recognition (**CVPR**), 2024.
11. Yang Ai, Xiaoqiang Zhou, **Huaibo Huang**, Lei Zhang, Ran He. Uncertainty-Aware Source-Free Adaptive Image Super-Resolution with Wavelet Augmentation Transformer. Computer Vision and Pattern Recognition (**CVPR**), 2024.
12. Qihang Fan, **Huaibo Huang**, Mingrui Chen, Hongmin Liu, Ran He. RMT: Retentive Networks Meet Vision Transformers. Computer Vision and Pattern Recognition (**CVPR**), 2024.
13. Tingkai Liu, Yunzhe Tao, Haogeng Liu, Qihang Fan, Ding Zhou, **Huaibo Huang**, Ran He, Hongxia Yang. DeVAn: Dense Video Annotation for Video-Language Models. Association for Computational Linguistics (**ACL**), 2024.
14. Xing Cui, Zekun Li, Peipei Li, **Huaibo Huang**, Xuannan Liu, Zhaofeng He. InstaStyle: Inversion Noise of a Stylized Image is Secretly a Style Adviser. European Conference on Computer Vision (**ECCV**), 2024.
15. Jin Liu, **Huaibo Huang**, Jie Cao, Ran He. ZePo: Zero-Shot Portrait Stylization with Faster Sampling. ACM International Conference on Multimedia (**ACM MM**), 2024.
16. Xuannan Liu, Pei Pei Li, **Huaibo Huang**, Zekun Li, Xing Cui, jiahao.liang, lixiong Qin, Weihong Deng, Zhaofeng He. FKA-Owl: Advancing Multimodal Fake News Detection through Knowledge-Augmented LVLMS. ACM International Conference on Multimedia (**ACM MM**), 2024.
17. Zi Wang, **Huaibo Huang**, Aihua Zheng, Ran He. Heterogeneous Test-time Training for Multi-modal Person Re-identification. AAAI Conference on Artificial Intelligence (**AAAI**), 2024.
18. Hao Sun, Jie Cao, Zhida Zhang, Tao Wu, Kai Zhou, **Huaibo Huang**. Learning Fine-grained and Semantically Aware Mamba Representations for Tampered Text Detection in Images. Pattern Recognition and Computer Vision (**PRCV**), 2024. (Best Student Paper Award)
19. Qihang Fan, **Huaibo Huang**, Xiaoqiang Zhou, Ran He. Lightweight Vision Transformer with Bidirectional Interaction. Neural Information Processing Systems (**NeurIPS**), 2023.
20. Rui Wang, Pei Pei Li, **Huaibo Huang**, Chunshui Cao, Ran He, Zhaofeng He. Learning-to-Rank Meets Language: Boosting Language-Driven Ordering Alignment for Ordinal Classification. Neural Information Processing Systems (**NeurIPS**), 2023.
21. Xiaoqiang Zhou, **Huaibo Huang**, Zilei Wang, Jie Hu, Ran He, Tieniu Tan. MSRA-SR: Image Super-resolution Transformer with Multi-scale Shared Representation Acquisition. International Conference on Computer Vision (**ICCV**), 2023.
22. Peipei Li, Rui Wang, **Huaibo Huang**, Ran He, Zhaofeng He. Pluralistic Aging Diffusion Autoencoder. International Conference on Computer Vision (**ICCV**), 2023.
23. Gengyun Jia, **Huaibo Huang**, Chaoyou Fu, Ran He. Rethinking Image Cropping: Exploring Diverse Compositions from Global Views. Computer Vision and Pattern Recognition (**CVPR**), 2022.

24. Xin Xie, Yi Li, **Huaibo Huang**, Haiyan Fu, Wanwan Wang, Yanqing Guo. Artistic Style Discovery With Independent Components. Computer Vision and Pattern Recognition (**CVPR**), 2022.
25. Gege Gao, **Huaibo Huang**, Chaoyou Fu, Zhaoyang Li, Ran He. Information Bottleneck Disentanglement for Identity Swapping. Computer Vision and Pattern Recognition (**CVPR**), 2021.
26. Peipei Li, **Huaibo Huang**, Yibo Hu, Xiang Wu, Ran He, Zhenan Sun. Hierarchical Face Aging through Disentangled Latent Characteristics. European Conference on Computer Vision (**ECCV**), 2020. (Oral, Co-first author)
27. Jie Cao, **Huaibo Huang**, Yi Li, Jingtuo Liu, Ran He, Zhenan Sun. Informative Sample Mining Network for Multi-Domain Image-to-Image Translation. European Conference on Computer Vision (**ECCV**), 2020.
28. Hao Zhu, **Huaibo Huang**, Yi Li, Aihua Zheng, Ran He. Arbitrary Talking Face Generation via Attentional Audio-Visual Coherence Learning. International Joint Conference on Artificial Intelligence (**IJCAI**), 2020.
29. Xiang Wu, **Huaibo Huang**, Vishal Patel, Ran He, Zhenan Sun. Disentangled Variational Representation for Heterogeneous Face Recognition. AAAI Conference on Artificial Intelligence (**AAAI**), 2019.
30. Weikuo Guo, **Huaibo Huang**, Xiangwei Kong, Ran He. Learning Disentangled Representation for Cross-Modal Retrieval with Deep Mutual Information Estimation. ACM International Conference on Multimedia (**ACMMM**), 2019.
31. Chaoyou Fu, Xiang Wu, Yibo Hu, **Huaibo Huang**, Ran He. Dual Variational Generation for Low-Shot Heterogeneous Face Recognition. Neural Information Processing Systems (**NeurIPS**), 2019.
32. Rui Wang, **Huaibo Huang**, Xufeng Zhang, Jixin Ma, Aihua Zheng. A Novel Distance Learning for Elastic Cross Modal Audio-Visual Matching. Workshops: 2019 IEEE International Conference on Multimedia and Expo (**ICMEW**), 2019. (Best Student Paper Award)